

QuadChain: Proof-Carrying Context Compression and Verifiable Memory Routing for Multiagent LLM Systems

research paper

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Abstract

Modern agent systems do not just need shorter prompts; they need compressed memory that can be trusted, routed, audited, and rejected when unsafe. We introduce QuadChain, a proof-carrying context compression architecture for multiagent LLM systems. QuadChain couples extractive compression with explicit evidence obligations, answer-concept preservation, role-aware routing, omission manifests, Merkle-style commitments, and tamper-evident handoff validation. This reframes compression as a control plane: the output is not a summary, but a verifiable memory object that tells downstream agents what was preserved, what was deleted, and why the handoff is safe to use. On a coding-agent benchmark, QuadChain compresses 2,250 input tokens to 1,383, preserving 41/41 required evidence and 38/38 answer concepts. In the 4-agent workflow eval, routing reduces total sent context from 9,000 to 2,283 tokens while preserving all measured obligations. In the actual 1k–300k scale ladder, QuadChain measures 24 concrete role prompts, preserves 144/144 audited payload evidence checks, and saves 1,253,400 workflow tokens (67.43%). The frontier harness adds 2,880 paired benchmark rows across 20 cases, 12 methods, four budgets, and three seeds; receiver task eval saves 74.22% workflow tokens. Adversarial handoff and route attacks are rejected 4/4 times. These results support a stronger thesis: the next compression layer for agentic AI is verifiable compressed memory, not blind summarization.

1 Introduction

Prompt compression research shows that LLM inputs often contain redundancy and can be shortened without proportional loss of quality. LLMLingua uses coarse-to-fine compression and budget control [1]; LongLLMLingua targets long-context settings and position bias [2]; LLMLingua-2 reframes compression as extractive token classification trained from distilled labels [3]; Selective Context explores self-information as a context filtering signal [4].

Agentic systems add a second problem. Context is no longer consumed once. It is copied, transformed, summarized, routed, and handed between agents. A receiver agent may act on compressed memory produced by another agent. Without verification, the receiver cannot distinguish a faithful compression from one that dropped a critical error, id, source path, constraint, or security warning.

2 Related Work

QuadChain is complementary to prompt-compression systems. It does not attempt to beat model-based compression methods on universal compression ratio. Instead, it adds a verification and routing layer around compression: required evidence is explicit, preserved facts are measured, omitted ranges are declared, and multiagent receivers verify a handoff before trusting it.

On the agent trust side, ERC-8004 proposes identity, reputation, and validation registries for autonomous agents [6], and ERC-8126 defines verification interfaces for registered agents [7]. TEE-based

and zk-style verification work points toward independent attestations of agent execution [8]. QuadChain maps those ideas onto compressed memory: the registry anchor is not raw context, but a validation commitment to a compression certificate.

3 Method

QuadChain represents each compressed packet as four commitments:

1. **Source commitment:** input hash and packet metadata.
2. **Compression commitment:** output hash, token delta, compression ratio, and omission ranges.
3. **Proof commitment:** required evidence, answer concepts, missing items, and answer-readiness score.
4. **Anchor commitment:** Merkle root, handoff hash, verifier version hash, and registry receipt.

quadchain compression pipeline

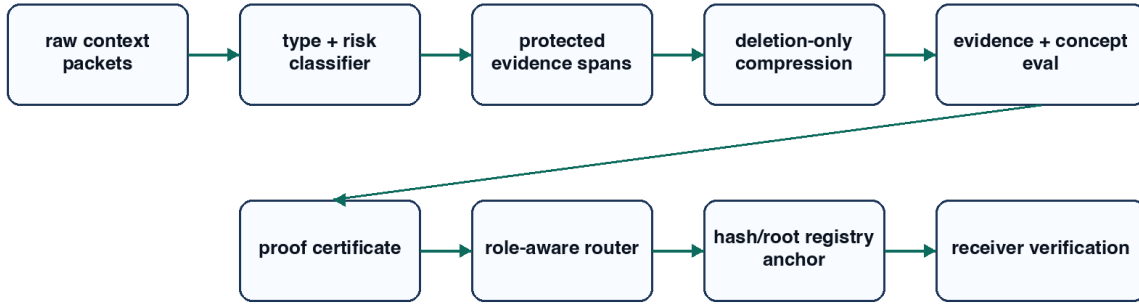


Figure 1: QuadChain compression pipeline.

3.1 Compression Policy

The current implementation is intentionally extractive and deterministic. It protects code-like spans, paths, ids, errors, dates, urls, key/value facts, and declared evidence. Low-signal lines such as repeated debug output, verbose chatter, and redundant logs are candidates for deletion. Role policies allow tool output to compress more aggressively than system or user intent.

3.2 Answer-Readiness Objective

Compressed packets are scored by exact preservation of required evidence and answer concepts:

$$\begin{aligned} \text{readiness} = & 0.65 \cdot \text{evidence} \\ & + 0.35 \cdot \text{concepts}. \end{aligned}$$

This metric is narrow but reproducible. It avoids relying on an external LLM judge during the hackathon and directly tests whether compressed context still contains the facts needed for downstream answers.

3.3 Multiagent Routing

The workflow eval compares a monolithic baseline, where every agent receives every raw packet, against QuadChain routing, where each role receives relevant compressed packets, certificate summaries, or aggregate proof material.

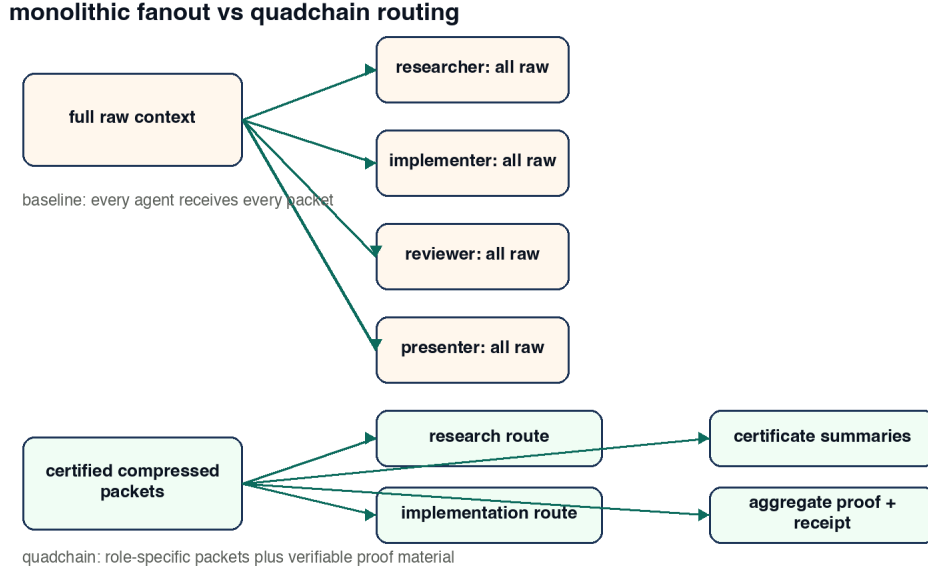


Figure 2: Monolithic fanout versus QuadChain role routing.

3.4 Handoff Verification

A receiver rejects a handoff if hashes, roots, verifier versions, registry commitments, required evidence, answer concepts, or route obligations fail. Raw context and evidence strings remain off-chain; only hashes and validation metadata are anchored.

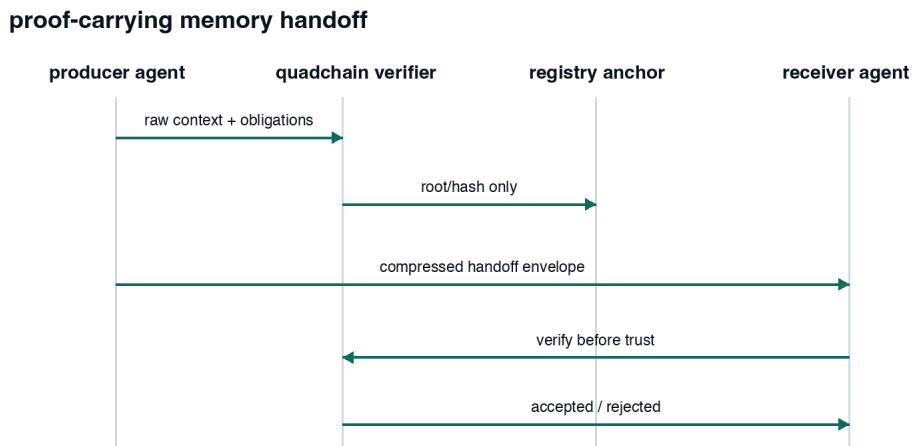


Figure 3: Proof-carrying memory handoff.

4 Experimental Setup

The evaluation uses five realistic coding-agent packets: an agent trace, failing test log, long agent history, noisy issue report, and research notes. Each packet has declared required evidence and answer concepts. Baselines compress to the same output-token budgets using head truncation, middle truncation, tail truncation, even-line sampling, and deterministic random-line deletion.

4.1 Frontier Benchmark Harness

The upgraded methodology emits one row per dataset adapter, item id, method, budget, seed, and scorer. The current run has 2,880 rows across 20 cases and 12 methods. Methods include raw context, QuadChain, head/middle/tail truncation, random deletion, keyword retrieval, embedding-top-k proxy, summary proxy, protected-span extraction, LLMingua-2 proxy, and Selective Context proxy. Budgets are 10%, 25%, 50%, and QuadChain-native, each repeated over three seeds.

The dataset adapters are `longbench_style_qa`, `needle_in_haystack_style`, `ruler_style_multihop`, `token_diet_local_fixture`. The nonlocal adapters are public-benchmark-style local slices inspired by `needle-in-haystack`, `RULER`, and `LongBench`-style tasks; they are explicitly labeled as adapters, not full external benchmark runs.

5 Results

5.1 Single-Context Compression

Table 1: Single-context compression result.

Input	Output	Saved	Reduction	Evidence	Concepts
2,250	1,383	867	38.53%	41/41	38/38

5.2 Baselines

Table 2: Same-budget baseline comparison.

System	Evidence	Concepts	Failure Rows
QuadChain / Token Diet	41/41	38/38	0
Best same-budget naive per packet	41/41	36/38	20/25

The stronger five-baseline set makes the comparison more honest: some naive routes preserve all evidence on easy packets, but the best aggregate naive strategy still loses answer concepts.

5.3 Multiagent Workflow

Table 3: Multiagent workflow routing result.

Raw	Routed	Saved	Reduction	Evidence	Concepts
9,000	2,283	6,717	74.63%	41/41	38/38

same-budget baseline failure surface

system	evidence	answer concepts	failure rows
quadchain	41/41	38/38	0 failure rows
best naive	41/41	36/38	20 failure rows

- headline: quadchain keeps exact evidence and answer concepts under same-budget pressure
- baselines include head, middle, tail, even-line, and deterministic random-line deletion
- failure rows mean at least one required evidence item or answer concept was lost

Figure 4: Same-budget baseline failure surface.

4-agent routing: actual token budget by role

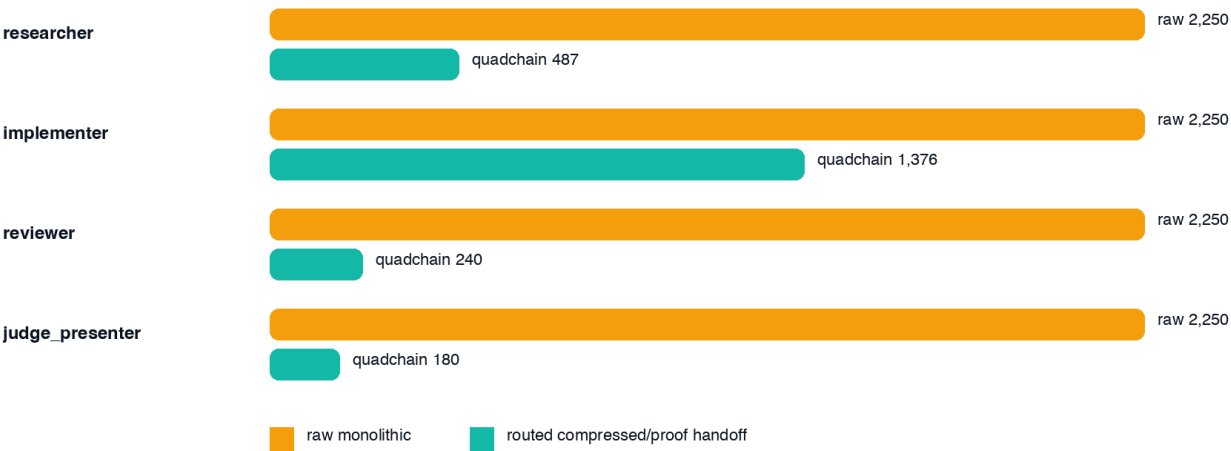


Figure 5: Actual per-role routing budget.

5.4 Adversarial Safety

Table 4: Handoff and route attack results.

Attack	Expected	Result
Tampered Merkle root	rejected	rejected
Dropped required evidence	rejected	rejected
Stale registry receipt	rejected	rejected
Invalid route missing role-critical packet	rejected	rejected

5.5 Measured Large Context

Table 5: Measured large-context compression.

Raw Context	Compressed	Saved	Reduction	Evidence
115,038	74,813	40,225	34.97%	12/12

5.6 Scale Ladder

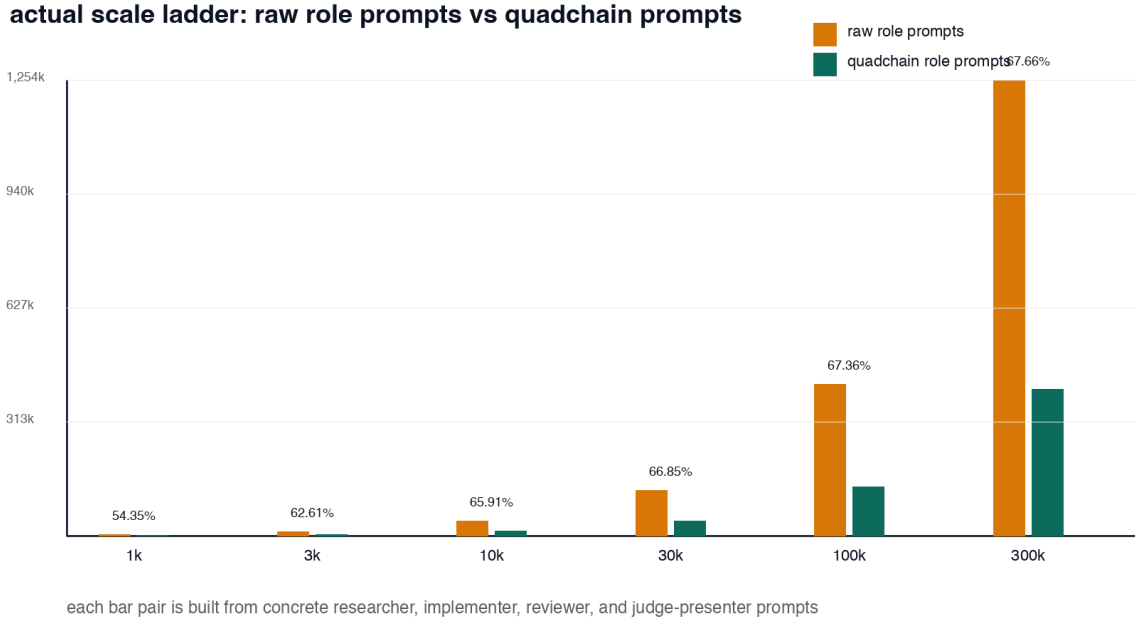


Figure 6: Actual scale ladder across 1k–300k-token generated traces.

Table 6: Multi-magnitude actual role-prompt scale ladder.

Magnitudes	Prompts	Smallest	Largest	Raw	QuadChain	Saved	Evidence
6	24	1,126	313,554	1,858,850	605,450	1,253,400	36/36

5.7 Frontier Benchmark Harness

The paired comparison result is intentionally mixed. QuadChain beats naive truncation and random deletion on the deterministic task score, while retrieval-style baselines can outperform it on some retrieval adapters. This is a feature of the methodology: QuadChain is a proof and routing layer, not a universal retrieval replacement.

Table 7: Per-magnitude actual routed prompt measurements.

Target	Input	Compressed	Raw Prompts	QuadChain Prompts	Saved	Reduction
1,000	1,126	765	4,659	2,127	2,532	54.35%
3,000	3,203	2,122	12,967	4,849	8,118	62.61%
10,000	10,534	6,909	42,291	14,419	27,872	65.91%
30,000	31,431	20,558	125,879	41,733	84,146	66.85%
100,000	104,632	68,030	418,683	136,664	282,019	67.36%
300,000	313,554	202,527	1,254,371	405,658	848,713	67.66%

Table 8: Frontier benchmark methodology result.

Metric	Value
Paired rows	2,880
Eval cases	20
Methods	12
QuadChain mean task score	0.6350
QuadChain CI95	[0.5987, 0.6782]
QuadChain mean token reduction	57.85%
Role payload evidence audit	144/144
Obligation leakage delta	0
Receiver raw task success	50.0%
Receiver QuadChain task success	75.0%
Receiver workflow reduction	74.22%

6 Discussion

QuadChain is strongest when context contains a mix of exact facts and low-signal noise. It is especially useful for coding agents because important details often look small: file paths, line numbers, error names, request ids, cookie attributes, dates, and one-line constraints. Those details should be protected, not summarized away.

7 Limitations

- Current public-style adapters are local deterministic slices, not full external LongBench, RULER, needle-in-haystack, SWE-bench, or GAIA runs.
- Local fixture obligations are hand-authored, though the frontier harness now separates local fixtures from public-style adapters.
- Exact evidence matching is reproducible but narrower than semantic task success.
- On-chain anchoring is simulated locally; the Solidity artifact is proof-of-design, not deployed infrastructure.
- The local compressor is deterministic and conservative; model-based compression could improve ratios.
- The scale ladder uses synthetic traces, but compression, role handoff prompts, token counts, and evidence checks are executed for every magnitude.

8 Future Work

- Replace or augment the local compressor with a learned keep/delete model trained on the generated policy dataset.
- Run LLM-as-judge and task-execution evals when API keys are available.
- Add ERC-8004-style validation ids and optional TEE or zkTLS attestation hashes to the registry contract.
- Replace local public-style adapters with full public long-context and coding-agent datasets.
- Support receiver-side selective rehydration when a route fails.

9 Conclusion

QuadChain reframes context compression as a verifiable systems primitive. The useful unit is not a shorter string; it is a compressed memory object with obligations, hashes, omission ranges, routing policy, and independent verification. This lets multiagent systems spend fewer tokens without blindly trusting compressed memory.

References

- [1] Jiang et al. *LLMLingua: Compressing Prompts for Accelerated Inference of Large Language Models*. arXiv:2310.05736. <https://arxiv.org/abs/2310.05736>
- [2] Jiang et al. *LongLLMLingua: Accelerating and Enhancing LLMs in Long Context Scenarios via Prompt Compression*. arXiv:2310.06839. <https://arxiv.org/abs/2310.06839>
- [3] Pan et al. *LLMLingua-2: Data Distillation for Efficient and Faithful Task-Agnostic Prompt Compression*. arXiv:2403.12968. <https://arxiv.org/abs/2403.12968>
- [4] Li. *Selective Context for LLMs*. https://github.com/liyucheng09/Selective_Context
- [5] Zhou et al. *Prompt Compression for Large Language Models: A Survey*. NAACL 2025. <https://aclanthology.org/2025.naacl-long.368.pdf>
- [6] Ethereum Improvement Proposals. *ERC-8004: Trustless Agents*. <https://eips.ethereum.org/EIPS/eip-8004>
- [7] Ethereum Improvement Proposals. *ERC-8126: AI Agent Verification*. <https://eips.ethereum.org/EIPS/eip-8126>
- [8] Eco. *TEEs for AI Agents: Verifiable Compute*. <https://eco.com/support/en/articles/14796365-tees-for-ai-agents-verifiable-compute>